



DIGITAL LIBRARY & LEARNING PLATFORM

System & Feature Design

A free, open-access digital library and learning platform for schools & colleges — course enrollment, an e-resource catalog, assignments, proctored tests, one-on-one doubt-clearing over chat & video, holistic student and course monitoring, and a course-grounded AI tutor that teaches from what each student is enrolled in.

100% free — no paid courses

Technology-agnostic design

Course-grounded AI tutor

Integrity-first assessments

Document: System & feature design (no technology stack) · **Audience:** Schools & colleges · **Roles:** Student · Instructor/Faculty · Librarian-Admin · Parent (optional)
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01 Design Goals & Principles

The platform is a **digital library** and **learning environment** in one: a place where students of a school or college browse and borrow reading material, enroll for free in structured courses, learn with an AI tutor grounded in their own coursework, prove their learning through assignments and proctored tests, and get one-on-one help from instructors — while instructors, administrators and (optionally) parents keep a clear, ethical view of progress.

OPEN BY DEFAULT

Every course and library resource is **free**. There are no prices, tiers, or subscriptions anywhere — access is governed by role and enrollment, never by payment.

GROUNDING & TRUSTWORTHY AI

The AI tutor teaches strictly from the enrolled course's material and **cites its sources**, so guidance is accurate and verifiable rather than open-ended.

INTEGRITY FIRST

Assessments are designed around fairness: proctoring, clear rules, and transparent violation handling protect the value of every result.

WHOLE-LEARNER VISIBILITY

Monitoring blends engagement, progress, assessment and help-seeking into a humane picture that highlights **who needs support**, not just scores.

HUMAN HELP, ONE TAP AWAY

The AI tutor handles routine questions; unresolved doubts escalate smoothly to a real instructor over chat or video.

SAFE FOR MINORS

Privacy, consent and an optional parent/guardian view are first-class design concerns for school-age users.

02 Actors & Roles

Four roles. The Parent/Guardian role is optional and intended for school deployments; colleges can disable it.

STUDENT The learner

- Browses the library, borrows e-resources, enrolls in courses (free)
- Learns via lessons, materials, and the course-grounded AI tutor
- Submits assignments; takes proctored tests
- Raises doubts and joins 1:1 chat/video sessions
- Tracks own progress, roadmaps, and certificates

INSTRUCTOR Faculty / teacher

- Authors courses, lessons, materials, assignments and tests
- Grades submissions; reviews proctoring flags
- Runs doubt-clearing sessions (chat & video)
- Monitors their courses and enrolled students
- Answers questions the AI tutor escalates

LIBRARIAN-ADMIN Administration

- Curates the e-resource catalog; manages borrowing rules
- Manages users, roles, classes/cohorts and enrollment policy
- Oversees institution-wide monitoring & integrity dashboards
- Configures the platform, notifications and academic calendar

PARENT Guardian (optional)

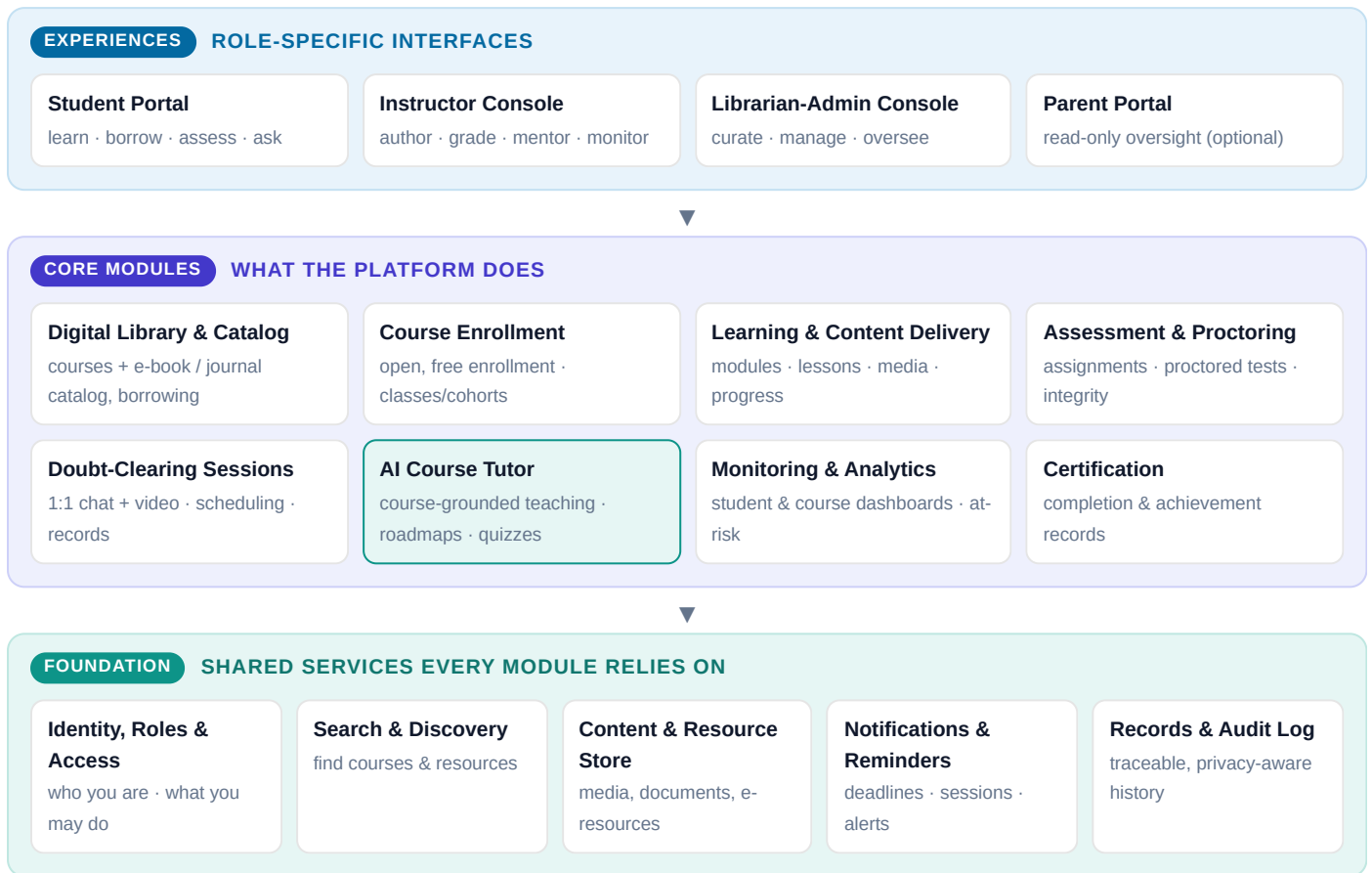
- Linked to one or more students (school deployments)
- Read-only view of progress, attendance & assessment summaries
- Receives notifications on milestones and at-risk alerts
- No access to raw content authoring or grading

03 System Building Blocks (module map)

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A conceptual map of the platform — described by capability, not technology. Experiences at the top consume core modules; every core module rests on a shared foundation.



Reading the map. Nothing in the experiences layer talks to the foundation directly — requests always flow through a core module that enforces the rules of that capability (e.g. only enrolled students reach a course's tutor; only an assignment's instructor sees its proctoring flags). This keeps **policy in one place per capability** and makes each module independently understandable.

Each module's purpose, key concepts, and the behaviour that defines it.

① Digital Library & Catalog

The "library" itself: two shelves under one roof.

- **Courses** — free, structured learning bundles (video, notes, downloadable materials)
- **E-resources** — a browsable catalog of e-books, journals & reference material
- **Borrow / read** — students open resources online or check them out for a set loan period; availability & returns are tracked
- Organised by **categories, subjects & tags**; full-text and metadata **search**
- Recommendations tie resources to the courses a student is enrolled in

② Course Enrollment

Open access — governed by role, never payment.

- **Self-enroll** in any open course; or be added to a **class/cohort** by an admin
- Optional **prerequisites** and **instructor approval** for restricted courses
- Enrollment unlocks course content, the AI tutor, and assessments
- Un-enroll / re-enroll supported; history retained for records
- No prices, carts, or subscriptions exist in the model

③ Learning & Content Delivery

How a course is structured and consumed.

- Course → **modules** → **lessons**; lessons carry video, text, and attached materials
- **Progress** is tracked per lesson and rolled up to a course percentage
- Resume-where-you-left-off; completion states drive certification
- Session recordings can be **transcribed & summarised** for review and accessibility

④ Assessment & Proctoring

Assignments and proctored tests — see §6 for the flow.

- **Assignments**: instructions, attachments, deadlines, late-policy, submission & grading with feedback
- **Proctored tests**: timed, with integrity monitoring and violation handling
- Attempts, marks and pass/fail are recorded and feed monitoring
- Results and feedback are returned to the student and (optionally) the parent

⑤ Doubt-Clearing Sessions

One-on-one human help — see §8 for the flow.

- Student **raises a doubt** (tied to a course/lesson) and picks **chat** or **video**
- Instant chat, or a **scheduled** video call with reminders
- Session transcript / notes are saved to the student's record
- Can be initiated directly, or as an **escalation from the AI tutor**

⑥ Certification

Recognising completion.

- Issued automatically when a course's completion & assessment criteria are met
- Verifiable achievement record tied to the student and course
- Visible to the student, instructor, admin and (optionally) parent

05 AI Course Tutor — the "internal Claude"

An assistant that behaves like a knowledgeable teacher for the specific courses a student is enrolled in — grounded in the real course material, not a generic chatbot.

WHAT IT DOES

- **Teaches from the course** — explains concepts using the enrolled course's lessons, notes & resources
- **Cites its sources** — every explanation points back to the lesson/resource it drew from, so students can verify and go deeper
- **Personal study roadmap** — generates and adapts an ordered path through the course based on the student's progress and gaps
- **Self-check quizzes** — creates short practice questions from the material and gives feedback on answers
- **Socratic help** — offers hints before answers to encourage genuine understanding

GUARDRAILS & BOUNDARIES

- **Stays in scope** — answers are bounded to courses the student is actually enrolled in
- **Knows its limits** — when material doesn't cover a question, it says so and can draw on general knowledge only if clearly flagged as beyond the course
- **Escalates to a human** — hands off to a doubt-clearing session with the instructor when a student is stuck or asks for it
- **No exam help during tests** — the tutor is unavailable inside a live proctored assessment
- **Age-appropriate & safe** — tone and content suited to the institution's learners

How a tutored answer is formed

- 1 **Student asks** a question from within a course they're enrolled in. The tutor knows which course, module and lesson they're on.
- 2 **Retrieve course context** — the tutor gathers the most relevant lessons, notes and resources *from that course*.
- 3 **Teach, grounded** — it composes an explanation from that material, adapted to the student's current progress, and **cites the source lessons**.
- 4 **Reinforce** — offers a next step on the personalized **roadmap** and, on request, a short **self-check quiz**.
- 5 **Escalate if needed** — if the question is out of scope or the student stays stuck, it offers to **book a doubt-clearing session** with the instructor.

Why "grounded" matters. Because the tutor teaches from the same material the student is graded on — and shows its citations — its guidance is consistent with the course, easy to trust, and safe to lean on. It also produces a valuable by-product for instructors: the questions students ask most reveal exactly where the course material is unclear (fed into Course Monitoring, §9).

06 Key Journey — Learn, Assess, Certify

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The student's end-to-end path. Monitoring (§9) observes every stage; the AI tutor (§5) is available throughout except inside a live test.



07 Proctored Test — integrity flow

Timed assessment with monitoring and a fair, transparent response to suspected violations.

- 1 **Pre-flight** — the student sees the rules, duration and what will be monitored, and gives informed consent to start.
- 2 **Locked session begins** — a timed attempt opens; the AI tutor and other aids are disabled for the duration.
- 3 **Integrity signals** — the session watches for defined events (e.g. leaving the test, switching context, copy/paste attempts) and records each with a timestamp.
- 4 **Graduated response** — minor events raise a visible warning to the student; repeated or serious events increment a violation count and can pause or end the attempt per policy.
- 5 **Auto-submit** — on time-out or completion the attempt is submitted with its answers and full integrity log.
- 6 **Human review** — the instructor sees the score alongside the flags and makes the final, accountable decision; the student is told the outcome and basis.

Fairness by design. Monitoring *informs* a human decision — it never silently fails a student. Rules are shown up front, flags are visible, and every consequential judgement is made by the instructor with the evidence in view. Repeat-offense limits and cool-down/lockout windows are configurable per assessment.

08 One-on-One Doubt-Clearing — flow

Human help over chat or video, reachable directly or as an escalation from the AI tutor.

CHAT SESSION (INSTANT)

- Student raises a doubt tied to a course/lesson
- Routed to an available instructor for that course; joins a queue if busy
- Real-time 1:1 messaging; can share snippets & resources
- Transcript saved to the student's record and course thread

VIDEO SESSION (SCHEDULED)

- Student requests a slot; instructor confirms a time
- Reminders sent to both; a private meeting link is generated
- 1:1 video call for deeper, face-to-face explanation
- Notes / recording & outcome logged after the call

- 1 Raise** — student describes the doubt (auto-linked to the current course & lesson) and picks chat or video.
- 2 Match & schedule** — routed to a course instructor; chat starts immediately, or a video slot is confirmed with reminders.
- 3 Resolve** — the session happens 1:1; shared context and resources keep it focused.
- 4 Record & close** — transcript/notes and a resolution status are saved; the doubt feeds monitoring (topics, response time, resolution rate).

Escalation path. When the AI tutor can't fully help, it offers a one-tap handoff that pre-fills the doubt with the conversation context — so the student never re-explains, and the instructor arrives already informed.

09 Monitoring & Analytics

Two lenses on the same activity: how each **student** is doing, and how each **course** is performing.

STUDENT MONITORING

- **Engagement** — logins, active time, lessons opened, resources read/borrowed
- **Progress** — lesson & course completion, roadmap position
- **Assessment** — assignment & test scores, attempts, integrity flags
- **Help-seeking** — doubts raised, sessions attended, tutor usage
- **At-risk signal** — a blended indicator flags students who need support early
- Parent-visible summary (optional): progress, attendance, milestones, alerts

COURSE MONITORING

- **Reach** — enrollments, active learners, completion rate
- **Outcomes** — score distributions, pass rates, assignment turnaround
- **Content usage** — most/least viewed lessons; drop-off points
- **Support load** — doubt-session volume & topics per course
- **Clarity gaps** — most-asked AI-tutor questions surface confusing material
- **Integrity** — proctoring-flag rates by assessment

10 Conceptual Data Model

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The entities the design revolves around and how they relate — expressed as concepts, independent of any storage technology.

Person name · contact has Role(s) {student-instructor- librarian-admin-parent}	Guardian Link Parent → Student oversight scope (school option)	Course title · subject free · open/restricted has Modules	Class / Cohort group of students grouped into Course (admin-managed)
Module in Course ordered has Lessons	Lesson in Module video · notes · media duration · order	Resource e-book · journal · ref catalog metadata borrowable	Loan / Read Student ↔ Resource borrowed · due · returned
Enrollment Student ↔ Course status · progress enrolled/completed	Progress Record per Enrollment/Lesson completion · last position time spent	Assignment in Course instructions · deadline marks · late-policy	Submission Student ↔ Assignment content · status marks · feedback
Test in Course timed · proctored rules · total marks	Test Attempt Student ↔ Test answers · score · passed attempt no.	Integrity Event per Test Attempt type · timestamp violation count	Certificate Student ↔ Course issued-on achievement record
Doubt Session Student ↔ Instructor mode: chat/video course/lesson · status	Session Message / Note in Doubt Session transcript · resolution	Tutor Conversation Student ↔ Course grounded messages cited sources	Roadmap / Quiz generated per Enrollment steps · questions

Relationships at a glance. A **Person** holds one or more **Roles**; a Parent links to Students via a **Guardian Link**. A **Course** contains **Modules → Lessons** and offers **Assignments & Tests**. A student's **Enrollment** is the spine that gathers their **Progress, Submissions, Test Attempts** (with **Integrity Events**), **Tutor Conversations**, generated **Roadmaps/Quizzes**, and any **Certificate**. **Resources** live in the catalog and are checked out via **Loan/Read** records. **Doubt Sessions** connect a student to an instructor around a specific course or lesson. Monitoring (§9) is a *view* computed over these entities — not a store of its own.

11 Access Control Matrix

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What each role may do. ● full · ▲ own/limited · ● none. "Own" means restricted to the user's own courses, students, or children.

CAPABILITY	STUDENT	INSTRUCTOR	LIBRARIAN-ADMIN	PARENT
Browse library & borrow resources	●	●	●	▲ view
Enroll in courses	●	●	▲ manage	●
Author courses / lessons / materials	●	●	▲ oversee	●
Use AI course tutor	●	▲ preview	●	●
Submit assignments / take tests	●	●	●	●
Grade & review proctoring flags	●	▲ own	●	●
Run doubt-clearing sessions	▲ request	●	●	●
View student monitoring	▲ self	▲ own	●	▲ child
View course monitoring	●	▲ own	●	●
Manage users, roles & catalog	●	●	●	●

12 Cross-Cutting Design Concerns

PRIVACY & MINORS

Data minimisation, clear consent, and parental oversight for school-age users. Monitoring is transparent to the student and framed for support, not surveillance.

ACCESSIBILITY

Captions & transcripts for video, readable typography, keyboard-friendly navigation, and low-bandwidth / offline reading for library resources.

ACADEMIC INTEGRITY

Rules shown up front, human-in-the-loop judgement, tutor disabled during tests, and full audit trails on every consequential action.

NOTIFICATIONS

Deadlines, session reminders, grade releases, and at-risk alerts — delivered per role with sensible defaults and opt-outs.

RELIABILITY OF LEARNING

Progress and submissions are never silently lost; auto-save and resume protect long lessons and timed attempts.

EXTENSIBILITY

Group study rooms, peer discussion, gamified streaks, multi-language tutoring, and institution-wide reporting are natural next steps.

In one line. A free, open digital library where students learn with a course-grounded AI tutor, prove their learning through fair proctored assessments, get one-on-one human help when they need it, and are supported early through humane monitoring — with instructors, admins and parents each given exactly the view their role calls for.